

C-1/W-1(Q-1)

moxug 7my

22/06/26

"Books" ~ sim

"Very Easy" ~ sim

Chap - 1 (Quiz - 1)

Data Mining: Extracting important (and interesting) ^{pattern on} information from any data set

Describe the essential steps of KDD:

* Knowledge Discovery from Data (KDD):

Extracting knowledge from data (synonym of Data Mining)

Steps (sequential):

(i) → Data Cleaning: to remove noise and inconsistent data

(ii) → Data Integration: ^{where} multiple data sources may be combined
↳ fixing format may be needed

(iii) → Data Selection: data relevant to the analysis are retrieved (and discarded what may not be needed)

(iv) → Data Transformation: data are transformed and consolidated into specific forms using (summary and aggregation) → ^{format}

(v) → Data Mining: intelligent methods are applied to extract data patterns

(vi) Pattern Evaluation: identify which patterns are more interesting

(vii) Knowledge Presentation: visualization and representation of findings

(ch-5)

[] Data Warehouse: NOT Database

~~"A"~~ ~~sto~~ A huge ~~storage~~ repository of data
of a large timeline (DB is short time) ↳ say server

C-2, W-1 (Q-1)

24/06/26

What kinds of patterns can be mined

*Characterization and discrimination:

Characterization → similar characteristics between data
→ identifying common characteristics between groups

Discrimination → comparison between two different groups

Mining frequent patterns, associations and correlations

repetitive pattern in a relative time/instances

foreshadow

connection between items



quantitative strength of association (-1 to 1) *ch. close enough, welcome back statistics*

support → % of ~~the~~ analysis among all that supports this case
confidence → chance of association (consider left)

✓ *frequent itemset
↳ set of frequent items
↳ same transaction purchase

*frequent subsequence
↳ set of frequent items in different transaction

a frequent substructure
↳ structure of frequent items

support will always be low since there will be a lot of transactions. Meanwhile, confidences are basically probability

support means how many times an association comes up
confidence is how many times among those associations this occurs.

C-3, W-1 (Q-1)

25/06/26

* Class Attribute $\rightarrow$ Feature/Attr that characterizes

* Training $\rightarrow$ has class attr. value

Testing $\rightarrow$ does not have class attr. value

* Outlier $\rightarrow$ Extreme values $\rightarrow$ foreshadow

Chap - 2

Getting to Know Your Data

Data Object and Attribute Types:

* Data Object: a data entity $\rightarrow$ instances

* Attribute: characteristics of a data object (column/feature)
 $\rightarrow$ called dimension in ~~DATA~~ DM

* Set of ^{all} ~~Attributes~~ ^{name} ~~Attributes~~ ^{name} called Attribute Vectors

$\rightarrow$ Nominal Attribute: some sort of name or category (but not ordered)

$\rightarrow$ Binary Attribute: any attribute with two values

$\rightarrow$ Ordinal Attribute: can be ranked $\rightarrow$ (high, medium, low) etc
 $\rightarrow$ required values have high magnitude

$\rightarrow$ Numeric

→ Numeric Attributes : Quantitative attributes (numbers)

→ Interval-Scaled; i) Does not have any true 0 point → 0 $\neq$ empty

→ Ratio-Scaled; i) True 0 point

02 C-4, W-2 (Q-2)

02/07/26

49 Central Tendency: which values are more "recurring"

→ Mean, Median, Mode

* Mean: Basic mean / Arithmetic mean =

$$\text{Weighted Mean, } \bar{x} = \frac{\sum w_i x_i}{\sum w_i}$$

* Normal Median

* Median: i) Sort (low to high)

ii) Count if (odd or even) → for even:
if N is numbers of data,
then take,

$$\frac{N-2}{2} \text{ and } \left(\frac{N-2}{2} + 1\right)$$

and take their average

* Interval Median: for binned data (

$$\text{median} = L_i + \left(\frac{\frac{N}{2} - (\sum \text{freq})_L}{\text{freq}_{\text{median}}} \right) \times \text{width} ; L_i = \text{Lower bound of median interval}$$

↳ ascending sorted

median interval: class that contains the $\frac{N}{2}$ th value

N = numbers of values of entire data set

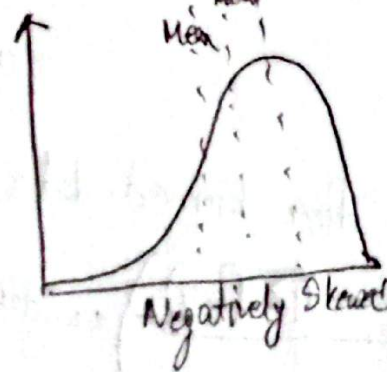
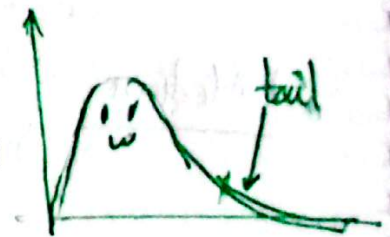
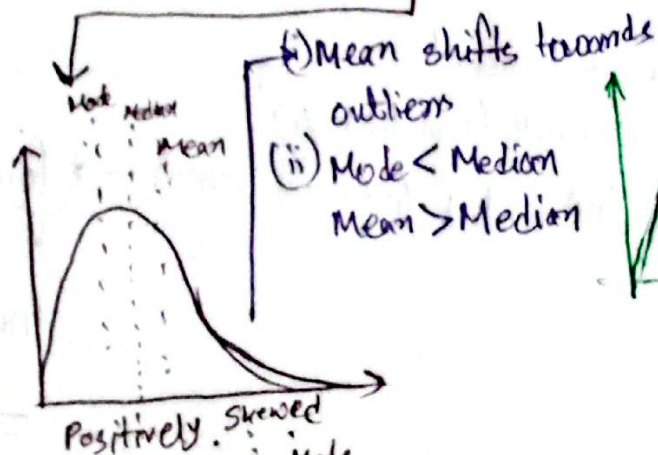
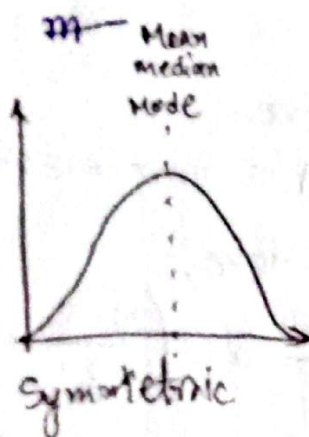
$(\sum \text{freq})_L$ = Cum. frequency upto median interval
width = range of frequency interval

* Mode: Numbers which appears most

→ for unimodal numeric data

moderately skewed (asymmetric)

$$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$$



C-5, W-3 (Q-2)

08/07/26

Dispersion of Data: -85

*Range $\rightarrow$ Difference betⁿ max and min values

*Quantile $\rightarrow$ Make equal parts ~~between~~ based on data points
after regular interval

$\rightarrow$ 2-Quantile $\rightarrow$ the datapoint that creates two ^{equal} halves of the whole dataset $\rightarrow$ median $\rightarrow$ check the formula for finding quantiles $\rightarrow$ HW
 $\hookrightarrow$ we usually discard the quantile

$\rightarrow$ 4-Quantile $\rightarrow$ the 3 datapoints that create 4 parts
 Q_1, Q_2, Q_3 Quantiles

↓
finds median

↓
median of the two parts

$\rightarrow$ 100-Quantile $\rightarrow$ makes percentile

$\therefore Q_1 = 25^{\text{th}}$ percentile

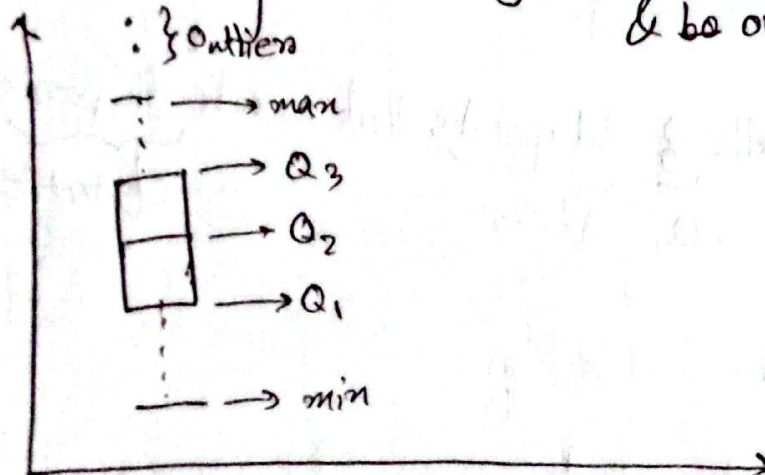
$Q_2 = 50^{\text{th}}$ "

$Q_3 = 75^{\text{th}}$ "

* Interquartile range $\rightarrow$
IQR

- (i) $x \leq Q_1 - (1.5 \text{ IQR}) \rightarrow$ ^{Lower} ~~Small~~ Outlier $\rightarrow$ -ve Outlier
(ii) $x \geq Q_3 + (1.5 \text{ IQR}) \rightarrow$ Large Outlier $\rightarrow$ +ve Outlier

* Box plot: Graphical Representation of using five numbers
(theory summary) $\rightarrow$ min, Q_1 , Q_2 , Q_3 , max
& be outliers



* Variance $\rightarrow$ ^{squared} avg. distance of points from mean
^{magnitude}
^{remove -ve}

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2$$

* Standard Deviation $\rightarrow \sigma$

$\rightarrow$ avg distance - - -

$\rightarrow$ use N for whole population
 $N-1$ " specific part
 $\downarrow$
we do this to avoid bias

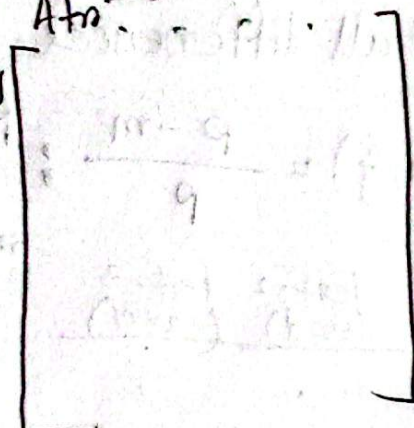
C-6, W-3(Q-2)

09/07/26

Data Matrix vs Dissimilarity Matrix (2.4.1)

(67, 104)

* Data Matrix : obj $\downarrow$ Attr $\rightarrow$

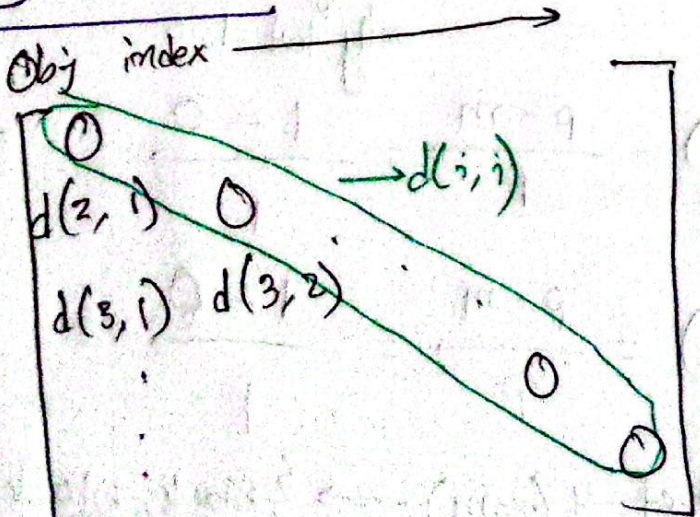


Collection of
 $\rightarrow$ Objects in ~~an~~ is data samples or feature vectors,
 $\downarrow$
vectors of attr
of obj

* Dissimilarity Matrix :

Obj index $\rightarrow$

Obj ind $\downarrow$



$d(i, j)$
 $\downarrow$
proximity measurement

Proximity Measure: → differently calculated for different data types

↳ take avg of all differences in attributes

* Nominal: $d(i, j) = \frac{p - m}{p}$; p = total number of nominal attribute
 m = match of i, j among nominal attributes

Say,

Obj identifier	test-1 (nom)	test-2 (ord)	test-3 (num)
1	code A	.	.
2	B	.	.
3	C	.	.
4	A	.	.

wait ???

$$d(1, 2) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

both none are same code (nominal)
 only test-1 is

$$d(2, 4) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

$$d(3, 1) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

lets add test-4 (nom) → { black, black, white, white }

$$d(1, 2) = \frac{2 - 1}{2} = 0.5$$

only black, black are same

* Binary:

→ symmetric: both values are equally important

→ asymmetric: one is more important
 → one is more useful for data analysis
 → the other might be more common

Contingency Table:
 (Symmetric)

	Object j		sum
	1	0	
Object i	1	q	r
	0	s	t
sum	$q+s$	$r+t$	p

q = No. of attr with 1

r = i has 1
 j has 0

s = i has 0
 j " 1

total ~~see~~ symmetric attr.

$$\therefore d(i, j) = \frac{r+s}{r+q+s+t} = \frac{r+s}{p}$$

For Asymmetric:

$$\therefore d(i, j) = \frac{r+s}{q+r+s} = \frac{r+s}{p-t}$$

we ignore 00

$\therefore$ for binary $\rightarrow$ avg (sym, asym)

(online)

C-7 (W-4?) Q-2

15/07/26



* Numeric

→ Euclidean $\rightarrow d(i, j) = \sqrt{(x_{i_1} - x_{j_1})^2 + (x_{i_2} - x_{j_2})^2 + \dots}$

→ Manhattan $\rightarrow d(i, j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \dots$

→ Minkowski $\rightarrow d(i, j) = \sqrt[h]{|x_{i_1} - x_{j_1}|^h + |x_{i_2} - x_{j_2}|^h + \dots}$

↳ generalizing euclidean & manhattan
usually $h = 1, 2$

→ Supremum $\rightarrow d(i, j)$

↓
the attr before
which dist will
be max

↓
for $x_1(1, 2)$ and $x_2(2, 5)$

$$\therefore \text{supremum} = |2 - 5| = 3$$

* Ordinal → i) String to numeric

ii) Normalize $\rightarrow z_{if} = \frac{x_{if} - 1}{M_f - 1}$; $M_f = \text{max value total number of unique value}$

iii) Use numeric → but for multiple ordinal

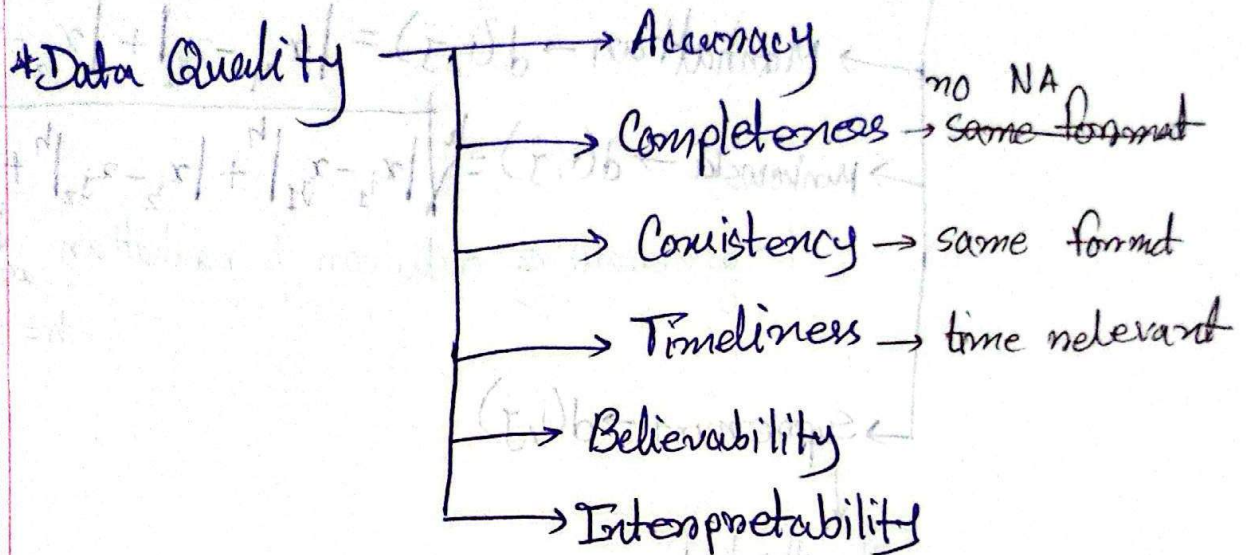
↳ use manhattan minkowski

Quiz - 2 on
week - 6 Wed
29th → ch - 2

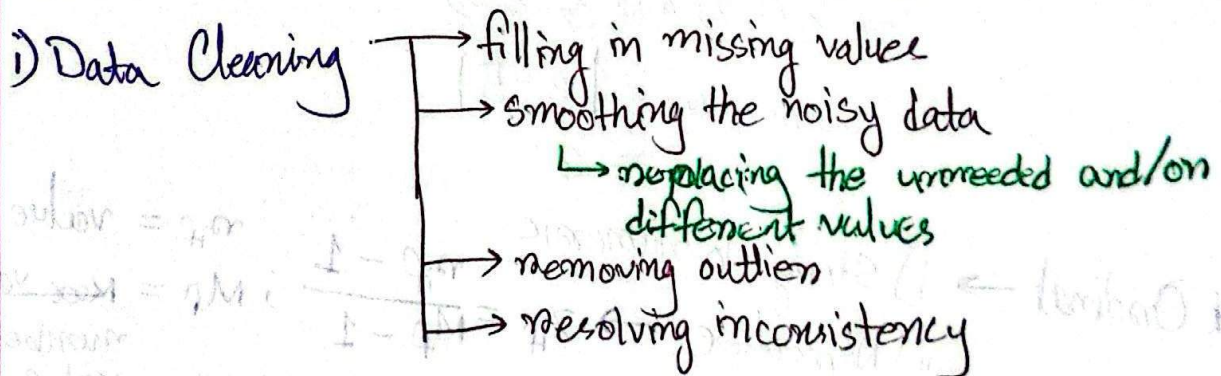
online
C-8, W-4? (a-2.5)

16/07/28

Data Preprocessing

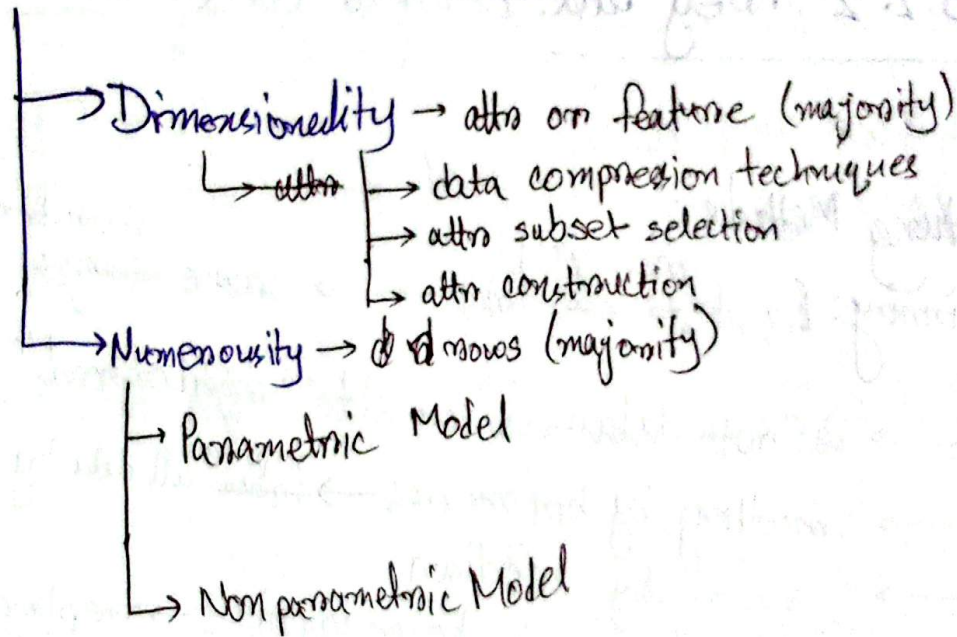


Major Tasks in Data Preprocessing:



ii) Data Integration → merging datasets

iii) Data Transformation Reduction



Data Cleaning:

i) Missing Value:

- i) Ignore the tuple / record ← highly inefficient
 - ii) Filling in manually ← only done when multiple / majority are missing
 - iii) Using a global constant
 - iv) Use mean or median of for all sampling samples belonging to the same class as given tuple
→ find the val not the entire attr. just the ones with same class / label
 - v) Use the most probable value
 - vi) Using mean and median
- Biases the data {

C-9, W-5 (Q-2.5)

20/07/26

3.2.2. Noisy Data: Deviated Data

↳ we smooth it

Smoothing Methods:

i) Binning: Equal ^{frequency} sized bins → same ^{number} range

↳ Sort data → consult "neighborhood"
↳ Smoothing by bin means → replace ^{mean} all data by the mean
↳ " by median
↳ " bin boundaries → replace all data by boundary values

* in exam we replace all values

ii) Regression: Find the "best" possible value to recalculate
Play: don't tell me I need to explain what regression is

iii) Outlier Analysis: [from before]

C-10, W-5 (Q-2.5)

22/07/26

3.3.2. Redundancy and Correlation Analysis:

if two attr are very strongly correlated $\rightarrow$ we will be able to retrieve one from another

* Correlation Analysis $\rightarrow$
 $\left\{ \begin{array}{l} \text{Nominal} \\ \text{Numerical} \end{array} \right.$

Nominal: String / Categorical $\rightarrow$ we use χ^2 test (chi square)
 $\left\{ \begin{array}{l} \text{Hypothesis test} \\ \text{Null hypothesis test} \end{array} \right.$
 Assuming no correlation $\rightarrow$ freq if not correlated

		Attr. 1		Total	
		catag 1	catag 2		
Attr. 2	catag 1	— ()	— ()		$\rightarrow$ expected freq (e_{ij})
	catag 2	— ()	— ()		$\rightarrow$ observed freq (o_{ij})
Total					

$$e_{ij} = \frac{\text{count}(i) \times \text{count}(j)}{n} ; n = \text{total}$$

$$\chi^2 = \sum \frac{(o_{ij} - e_{ij})^2}{e_{ij}} \rightarrow \text{if } \chi^2 > \text{Table} \rightarrow \text{then NULL Hypothesis then false} \rightarrow \text{correlation exists}$$

$$\text{degree of freedom} = (\text{row} - 1)(\text{col} - 1)$$

significance value $\rightarrow$ given in ques

	Sig value
Deg of freedom	

C-11, W-5 (Q-2.5)

23/07/26

* Significance Level $\rightarrow$ Prob of making mistake if Null Hypothesis fails

* Correlation $\rightarrow$ Strong if $\chi^2 \gg \text{table}$
Weak if $\chi^2 > \text{table}$

* Deg of Freedom $\rightarrow$ How many values can be independent
 $\rightarrow$ no significance in this regard as we take values from dataset anyways

Numerical: Pearson's Product Moment Coefficient

$\rightarrow$ Covariance $\rightarrow$ take avg $\rightarrow x - \bar{x}, y - \bar{y} \rightarrow$ if same sign then relation
 $\rightarrow$ Correlation Standard Deviation

* Covariance doesn't indicate intensity $\rightarrow$ sign indicates relation
Correlation can but magnitude doesn't mean anything

* We have concepts of r^2 and covariance:

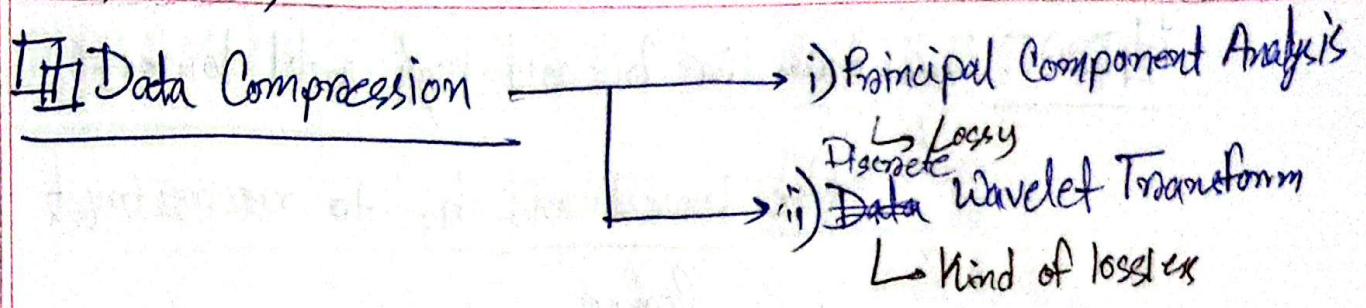
$$\sum \frac{(x - \mu_x)(y - \mu_y)}{\sigma_x \sigma_y}$$

$$\text{Cov} = \sum \frac{(x - \mu_x)(y - \mu_y)}{n \cdot \sigma_x \sigma_y}; \mu_i = \text{avg}(i \text{ col})$$

Q-12, Q-2.5, (w-6)

27/07/26

Data Compression



PCA: Using a single value to express the multiple attributes
↳ called Principal Component

* PC1 → Gives highest priority to the attr. with highest variance

↳ $n(\text{Attr.}) = n(\text{PC})$
but we keep 2
↳ but only one is kept

* PC2 → [yeah I think you can make we assign weight corresponding to priority the rest out]

* PC3 → Gives ~~lowest~~ highest priority to the attr. with ~~lowest~~ ^{third} variance

* The more PC, the easier the to retrieve the original data. → but not entirely

* Find λ → Use Eigen vectors → Use the values ^{called Loading} ~~equal~~ in vectors PC
↳ Eigen values
represents variance

DWT: Break down in low and high until lowest

↳ Use lowest and h_i to reconstruct
Coarse

C-13, W-6, Q-2.5

29/07/26

PCA Math:

<u>Stu Id</u>	<u>Hours Studied</u>	<u>Exam Marks</u>
1	2 (-1.26)	1 (-1.26)
2	3 (0.63)	2 (-0.63)
3	4 (0)	3 (0)
4	5 (0.63)	4 (0.63)
5	6 (1.26)	5 (1.26)

z-score
Step-1: Normalization

$$z = \frac{x - \bar{x}}{\sigma_x}$$

Step-2: i) Covariance using normalization
 ii) " " matrix C

$$\text{CoV} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= 0.992 \approx 1$$

$$C = \begin{bmatrix} \text{Var}(x) & \text{CoV}(x, y) \\ \text{CoV}(y, x) & \text{Var}(y) \end{bmatrix}$$

Step-3: Eigen Values λ

$$C \cdot \vec{v} = \lambda \vec{v} \rightarrow C \vec{v} = \lambda \vec{I} \vec{v}$$

$$\text{or, } C \vec{v} - \lambda \vec{v} = 0$$

$$\therefore \vec{v} (C - \lambda) = 0$$

$$\therefore \text{Either } \vec{v} = 0 \text{ or } C - \lambda \vec{I} = 0$$

But $\vec{v} = 0$ is not possible $\rightarrow$ loadings can never be 0

But $C - \lambda \vec{I} = 0$ is also not possible

but $|C - \lambda \vec{I}| = 0$ is possible

$$\text{or, } \begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$\text{or, } (1-\lambda)^2 - 1 = 0$$

$$\text{or, } 1 - 2\lambda + \lambda^2 - 1 = 0$$

$$\text{or, } \lambda^2 - 2\lambda = 0$$

$$\text{or, } \lambda(\lambda - 2) = 0$$

$$\therefore \boxed{\lambda = 0}, \boxed{\lambda = 2} \rightarrow \text{PC1}$$

$\downarrow$
PC2

Step-4: Eigen Vector

When $\lambda = 2$,

$$\vec{v}(C - \lambda I) = 0$$

$$\text{or, } \vec{v}(C - 2I) = 0$$

$$\text{or, } \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} -v_1 + v_2 & v_2 - v_1 \end{bmatrix} = 0$$

$$\text{or, } \begin{bmatrix} t & t \end{bmatrix} = 0$$

$$\therefore \|\vec{v}\| = \sqrt{2}t$$

$$\therefore \hat{v} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

C-14, W-7, Q-3

06/08/26

DWT Math

i) Set ~~low~~ must be expressible as 2^m

↳ if not, pad with 0s

ii) a) Avg b) Diff of pairs
 ↓ ↓
 ~~high~~ ~~low~~
 low high

iii) Step 2 (ii) of Avg₀ until 1 value

$$\# [8 \ 12 \ 16 \ 40 \ 45 \ 60, 80, 100]$$

$$\text{Avg}_1: \left[\frac{8+12}{2}, \frac{16+40}{2}, \frac{45+60}{2}, \frac{80+100}{2} \right] = [10, 28, 52.5, 90]$$

$$\text{Diff}_1: \left[\frac{8-12}{2}, \frac{16-40}{2}, \frac{45-60}{2}, \frac{80-100}{2} \right] = [-2, -12, -7.5, -10]$$

Again,

$$\text{Ag Avg}_2: \left[\frac{10+28}{2}, \frac{52.5+90}{2} \right] = [19, 71.25]$$

$$\text{Diff}_2: \left[\frac{10-28}{2}, \frac{52.5-90}{2} \right] = [-9, -18.75]$$

Again,

$$\text{Avg}_3 = \left[\frac{17+71.25}{2} \right] = [45.125]$$

$$\text{Diff}_3 = \left[\frac{17-71.25}{2} \right] = [-26.125]$$

$$\therefore \text{Coarse} = 45.125$$

$$\therefore \text{Required Vector} = [45.125, -26.125, -9, -18.75, -2, -12, -7.5, -10]$$

C-15, W-8, Q-3

17/08/26

Attribute Subset Selection: Removing redundant data
(Kind of ~~reduction~~)

- * Forward Elimination
 - * Backward Elimination
 - * Decision Tree Induction
1. Stepwise Forward

2. Stepwise Backward

3. Combination of Stepwise Forward and Backward

* Stepwise Forward:

$E = \{ \}$

$A = \{ A, B, C, D \}$

Take the highest % of measure from A with E

For ML

~~Insert~~ to E

Loop until combination ~~accuracy~~ measure is lowered

* Stepwise Backward:

$$A = \{A, B, C, D\}$$

Remove element from A and take % measure

Until % decreases

* Combination of Forward and Backward:

Take the Best single } For E
Remove " worst element }

→ This is because if A is Best ACD is ~~worst~~ ^{Best}
Then A can be selected and B can be removed
A will (probably) not be in ~~the~~ subset stepwise
backward

Sampling: Taking part of instances (rows)

→ ~~Random~~ Simple Random Sample Without Replacement (SRS WOR) → single entry once (cut)

→ Simple Random Sample With Replacement (SRSWR)
↳ multiple times (copy)

→ Cluster → Take all from a range → convert that some cluster maybe lost

→ Stratified → Take SRS with ratio to the original dataset

Data Cube Aggregation: Fixing values

- * Data Cube $\rightarrow$ ~~Data~~ DS of Data Warehouse
- Heart of Data Warehouse
- $\rightarrow$ Multi-Dimensional ~~Array~~ DS (usually array)
- $\rightarrow$ Fix
- $\rightarrow$ If some cannot be aggregated $\rightarrow$
 - $\rightarrow$ discard
 - $\rightarrow$ construct new

Data Transformation Strategies

Normalization: Getting fixed range

$$* \text{Min max normalization } (v_i) = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new max}_A - \text{new min}_A) + \text{new min}_A$$

$$* \text{Z-Score Normalization } (v_i) = \frac{v_i - \bar{A}}{\sigma_A}; \bar{A} = \text{average}$$

Ch-4: Data Warehouse **

Online Analytical Processing → OLAP → DW
" Transactional " → OLTP → DB

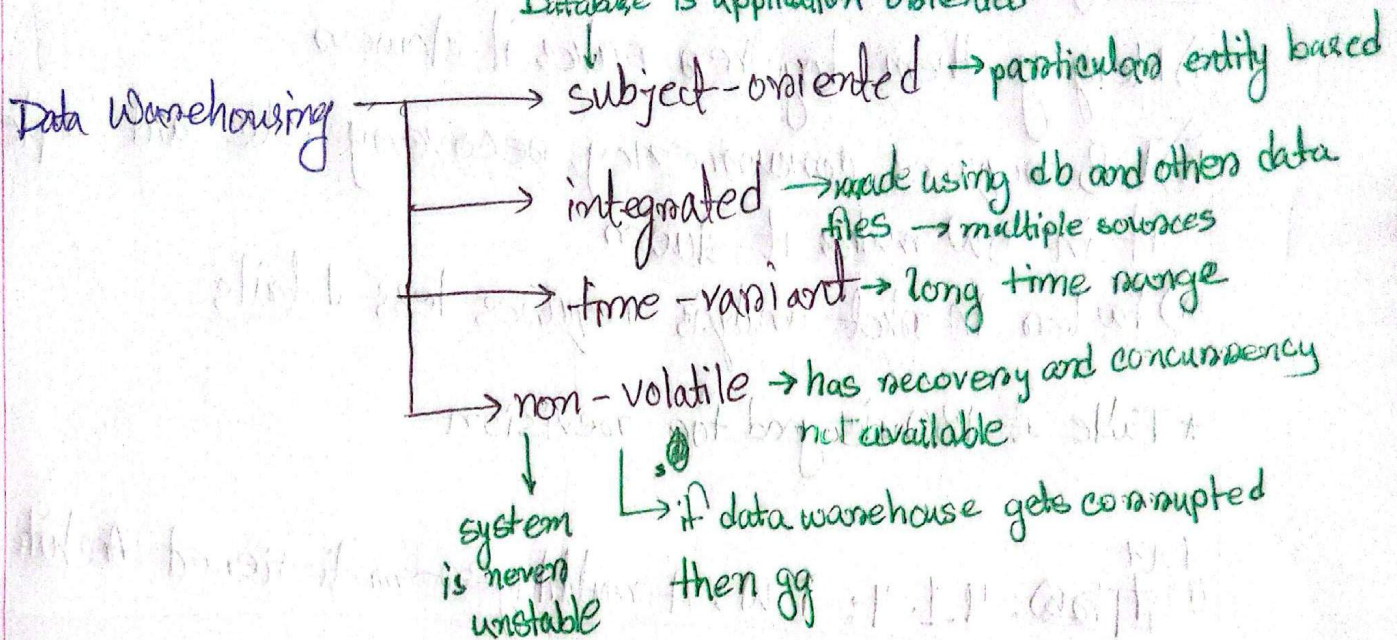
+ Data Warehouse is made using Database using Aggregation
+ DB has details | Data Warehouse has compressed details of DB

+ DW is used to analyze conditions

DB has recent but small data | DW has overall (old) but huge data

Data Warehousing

Database is application oriented



Differences between DB (OLTP) and DW (OLAP)

	(OLTP) DB	(OLAP) DW
i) users and system orientation	customer oriented	market oriented
ii) Data contents	Too detailed for decision-making	Short but huge and good for decision-making
iii) Design	Entity-Relation (Table)	Different schemes (Star, Snowflake, Constellation and Data cube)
iv) View	Current	Historic
v) Access Pattern	Read & Write	Read

27/08/26

Why Datawarehouse:

- i) Merging them together makes it slower
 - ii) DB requires concurrency, recovery etc and implementing it in DW makes it slower
 - iii) Faster market analysis requires less details
- * Table at 180 is good for revision

Q4. 4.1.4: DW: A multi-layer multi-tiered Architecture:

+ 3 tiers (but can be more)

↳ Most popular one

- * Take data from multiple heterogeneous sources

Frontend Tools

OLAP ^{↑ output} Servers
(cube)
↑
(aggregated)
data base

* Schemas
are more
detailed than
Cube

(Database) $\xrightarrow[\text{skipped}]{\text{also use}}$ DW Schema

Data Mart

Database Table

DW Servers

Multiple Servers
for a DW

Refresh
Load
Transform &
Clean
Extract

External Source